The book walkthrough: one command, then Excel

What the firm asked for, 14 Oct 2026: *"i want you to make it work so i can generate this workbook at work with the script, it should be debugged and walked through and tied out."*

This is the walk: the workbook built the way it is built at work, then used in Excel 16 the way a person uses it (clicking the buttons, typing in the yellow row, saving, closing and opening it again), and every answer read back from Excel and set beside an answer written down beforehand from the source files.

1. Build

From `threshold-engine/`:

```
python -m threshold_engine peers pull --cache books/ind_cache --first 2006Q1 --last 2026Q2 --products "Cre
dit card" Mortgage "Home equity" "Other consumer" Auto --delinquency yes --mergers 2005-01-01
python -m threshold_engine book books/consumer-public.json consumer-credit.xlsx
```

The walkthrough ran the second command on a copy of the book whose only change is where the cache is (`prepare.py`): the cache pulled on 14 Oct 2026 was used rather than pulled again. Expected: `Wrote consumer-credit.xlsx: 164 series, 143 quarters; open it on Summary, decide on Decide`. It took 21 min 40 s on the round 3 run of 19 Oct 2026, with the full test suite running beside it (8 min 48 s alone before round 3), almost all of it the peers build reading the cache.

2. The steps

The steps are cumulative, as a session is: what is typed in one step is still there in the next.

One set of lines per product and measure (the firm, 15 Oct 2026: *"we don't need to set thresholds on all 3, we need to use all 3 to set appropriate thresholds"*): Decide has 35 rows, and each row's **Lines set from** picks the source whose history sets its lines.

Step	What you do	What it shows
B1	Open the workbook	It opens on Summary; the tabs in round 3's order (Peers data, Peers delinquency data, Internal, Internal data, Assess, Chart and both workings tabs hidden); two slicers, one button each; Decide shows the first product and measure, no source picked, nothing decided; the counts, the Large banks strip, Needs your attention empty, the pack line on Peers, Decide and Signals, Reporting changes last on Sources; no links, no errors
B2	Read Summary and Lines by source	Every series Summary shows: its latest value, a dash where its rating goes; every series' own lines equal the reference; the backtest folded into Lines by source for two series, one too short to test
B3	On Decide, click Credit card and Nonaccrual, and pick KeyBank under Lines set from	One row; the lines are KeyBank's own; every source rated on them, beside what its own history would give; Thresholds holds the same
B4	Type 3.5 under How rare High is (z)	High's line moves for every source of the row; Thresholds and Summary follow
B5	Leave out 2012Q3 to 2013Q4, the period Decide suggests, with a reason	KeyBank's lines move as the reference says, the industry's own lines do not; Assess lists the period under KeyBank's series, whole
B6, B6b	Clear the reason; type it back	Refused, in words, on Decide and Thresholds; then the lines as in B5

Step	What you do	What it shows
B7	Accept these lines, with a note	Decided: 1 of 35; the decision on Thresholds, and on Lines by source beside KeyBank only
B8	Click Auto and Net charge-offs, and pick Industry (every bank)	KeyBank's life-to-date history rated on the lines the industry's longer history gives
B9	Click Mortgage and Net charge-offs, and pick Industry (every bank)	Refused, with the number that would fix it
B10	Save, close, open again	Every pick, decision and setting still there, and the lines with them; Summary's counts and Needs your attention
B11 to B13	On Peers: Regionals only; Every peer with JPMorgan off; JPMorgan back	The peer group's latest values are the composite over the banks left in, and the pack line names those switched off
B14	Open Signals	The controls and Picked, the pack line, where the dollars are, two fit tables

A step passes when every cell it reads says what `steps.json` says: a number within 1e-9, text exactly.

3. How it was run

```
python docs/walkthrough-book/prepare.py PATH-TO-CACHE
python -m threshold_engine book docs/walkthrough-book/out/book.json docs/walkthrough-book/out/consumer-cre
dit.xlsx
python docs/walkthrough-book/expected.py docs/walkthrough-book/out/book.json
copy docs\walkthrough-book\out\consumer-credit.xlsx docs\walkthrough-book\out\walk.xlsx
powershell -NoProfile -ExecutionPolicy Bypass -File docs\walkthrough-book\walk.ps1
python docs/walkthrough-book/compare.py
```

`expected.py` writes the answers from the series CSVs and `scale.py`, never from the workbook. `walk.ps1` does each step in Excel through Excel's own automation and reads each answer back. `compare.py` writes `RESULT.md`.

And, separately, that the file opens without Excel repairing it:

```
powershell -NoProfile -ExecutionPolicy Bypass -File tools\check_slicers_in_excel.ps1 -Path docs\walkthroug
h-book\out\consumer-credit.xlsx -Sheet Decide -Table Decisions -Shapes 3 -Expect "Slicer_Product=Credit ca
rd|Slicer_Measure=Net charge-offs" -SwitchCache Slicer_Measure -SwitchTo "Gross charge-offs"
```

"No errors" in B1 means none a reader can see: on the visible tabs, outside hidden rows and columns. The rest are counted by kind and are all #N/A, the gap a chart leaves where a quarter has no value (on the run of 15 Oct 2026: Assess 169, Chart 8,470, Decide 249, Peers workings 1,556).

4. What it does not prove

- The peer tabs are counted for errors and links (B1) but their numbers are not checked against a reference here; `tests/test_peers*.py` hold them.
- One series is tied to the filing (`../tie-out/keybank-card-nonaccrual-2026-10-14/`); the other 104 rest on the same code and the reference, not on a filing each.
- Three of the 35 rows have a source picked in the walk (Credit card nonaccrual, Auto and Mortgage net charge-offs); the other 32 are read blank (a dash) on Summary, and every source's own lines are read for all 105 on Lines by source.

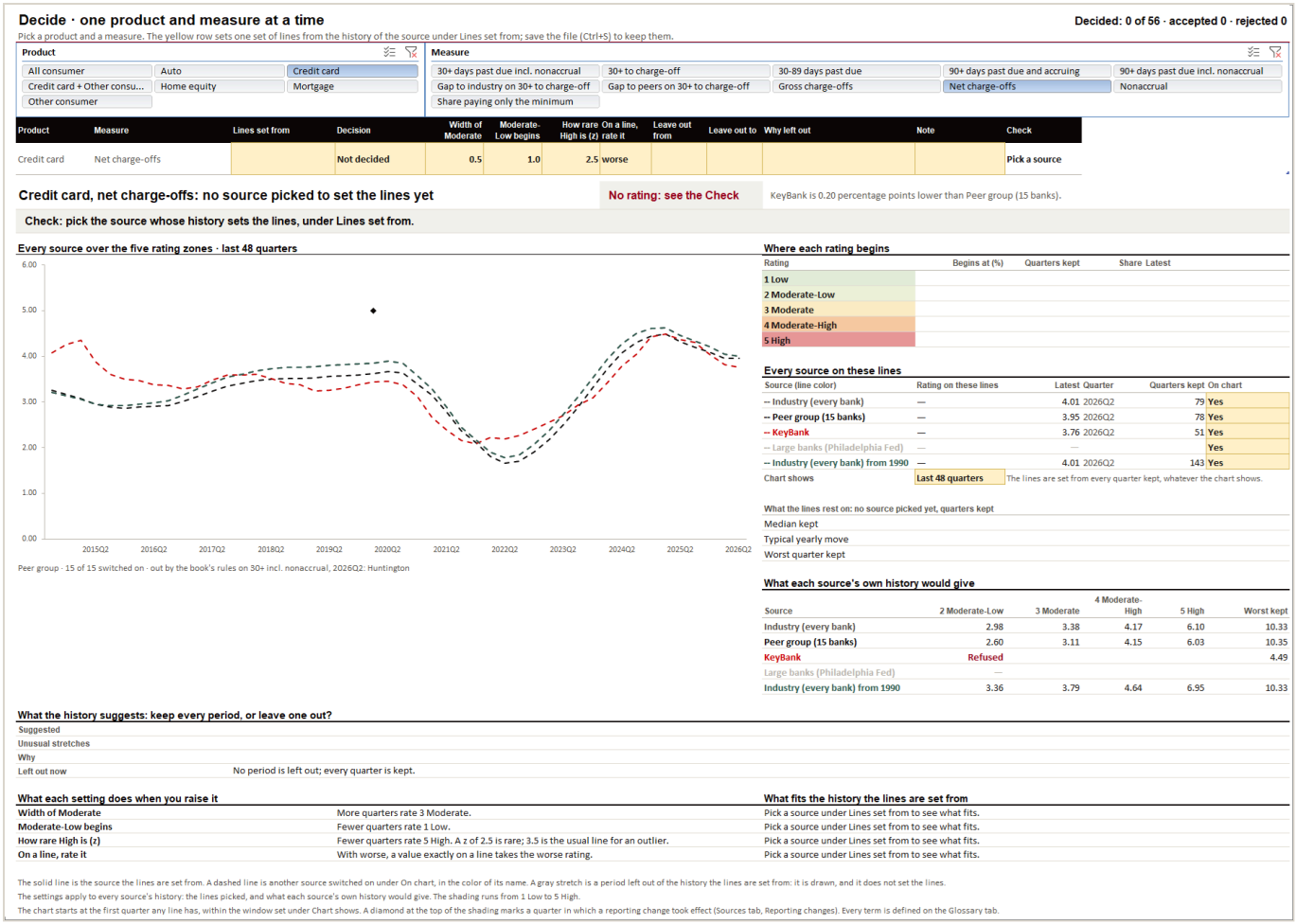
- Thresholds' latest value is written by the build, not a live formula on Data: pasting a corrected quarter into Data moves the lines but not the latest value (found by the tie-out).

5. The result

2502 of 2502 reads as expected.

Step	What you do	Reads as expected
B1	Open the workbook as built	47 of 47
B2	Read Summary, and every source's own lines on Lines by source	1972 of 1972
B3	On Decide, click Credit card and Nonaccrual, and pick KeyBank under Lines set from	59 of 59
B4	Type 3.5 under How rare High is (z), in the yellow row	55 of 55
B5	Leave out 2012Q3 to 2013Q4, the period Decide suggests, with a reason	41 of 41
B6	Clear the reason	3 of 3
B6b	Type the reason back	26 of 26
B7	Pick Accept these lines and write a note	4 of 4
B8	Click Auto and Net charge-offs, and pick Industry (every bank) under Lines set from	31 of 31
B9	Click Mortgage and Net charge-offs, and pick Industry (every bank)	25 of 25
B10	Save (Ctrl+S), close Excel, open the file again	39 of 39
B11	On Peers, set Banks in the pack to Regionals only	11 of 11
B12	Set Banks in the pack back to Every peer, and JPMorgan's cell to No	11 of 11
B13	Set JPMorgan's cell back to Pack	14 of 14
B14	Open Signals	164 of 164

6. What Excel showed



Decide · one product and measure at a time

Decided: 0 of 56 · accepted 0 · rejected 0

Pick a product and a measure. The yellow row sets one set of lines from the history of the source under Lines set from; save the file (Ctrl+S) to keep them.

Product

All consumer | Auto | Credit card
Credit card + Other consu... | Home equity | Mortgage
Other consumer

Measure

30+ days past due incl. nonaccrual | 30+ to charge-off | 30-89 days past due | 90+ days past due and accruing | 90+ days past due incl. nonaccrual
Gap to industry on 30+ to charge-off | Gap to peers on 30+ to charge-off | Gross charge-offs | Net charge-offs | Nonaccrual
Share paying only the minimum

Product	Measure	Lines set from	Decision	Width of Moderate	Moderate-Low begins	How rare High is (z)	On a line, rate it	Leave out from	Leave out to	Why left out	Note	Check
Credit card	Net charge-offs		Not decided	0.5	1.0	2.5	worse					Pick a source

Credit card, net charge-offs: no source picked to set the lines yet

No rating: see the Check

KeyBank is 0.20 percentage points lower than Peer group (15 banks).

Check: pick the source whose history sets the lines, under Lines set from.

Every source over the five rating zones · last 48 quarters

Peer group · 15 of 15 switched on · out by the book's rules on 30+ incl. nonaccrual, 2026Q2: Huntington

Where each rating begins

Rating	Begins at (%)	Quarters kept	Share Latest
1 Low			
2 Moderate-Low			
3 Moderate			
4 Moderate-High			
5 High			

Every source on these lines

Source (line color)	Rating on these lines	Latest Quarter	Quarters kept	On chart
-- Industry (every bank)	—	4.01 2026Q2	79	Yes
-- Peer group (15 banks)	—	3.95 2026Q2	78	Yes
-- KeyBank	—	3.76 2026Q2	51	Yes
-- Large banks (Philadelphia Fed)	—	—		Yes
-- Industry (every bank) from 1990	—	4.01 2026Q2	143	Yes

Chart shows: Last 48 quarters — The lines are set from every quarter kept, whatever the chart shows.

What the lines rest on: no source picked yet, quarters kept

Median kept
Typical yearly move
Worst quarter kept

What each source's own history would give

Source	2 Moderate-Low	3 Moderate	4 Moderate-High	5 High	Worst kept
Industry (every bank)	2.98	3.38	4.17	6.10	10.33
Peer group (15 banks)	2.60	3.11	4.15	6.03	10.35
KeyBank	Refused				4.49
Large banks (Philadelphia Fed)	—				
Industry (every bank) from 1990	3.36	3.79	4.64	6.95	10.33

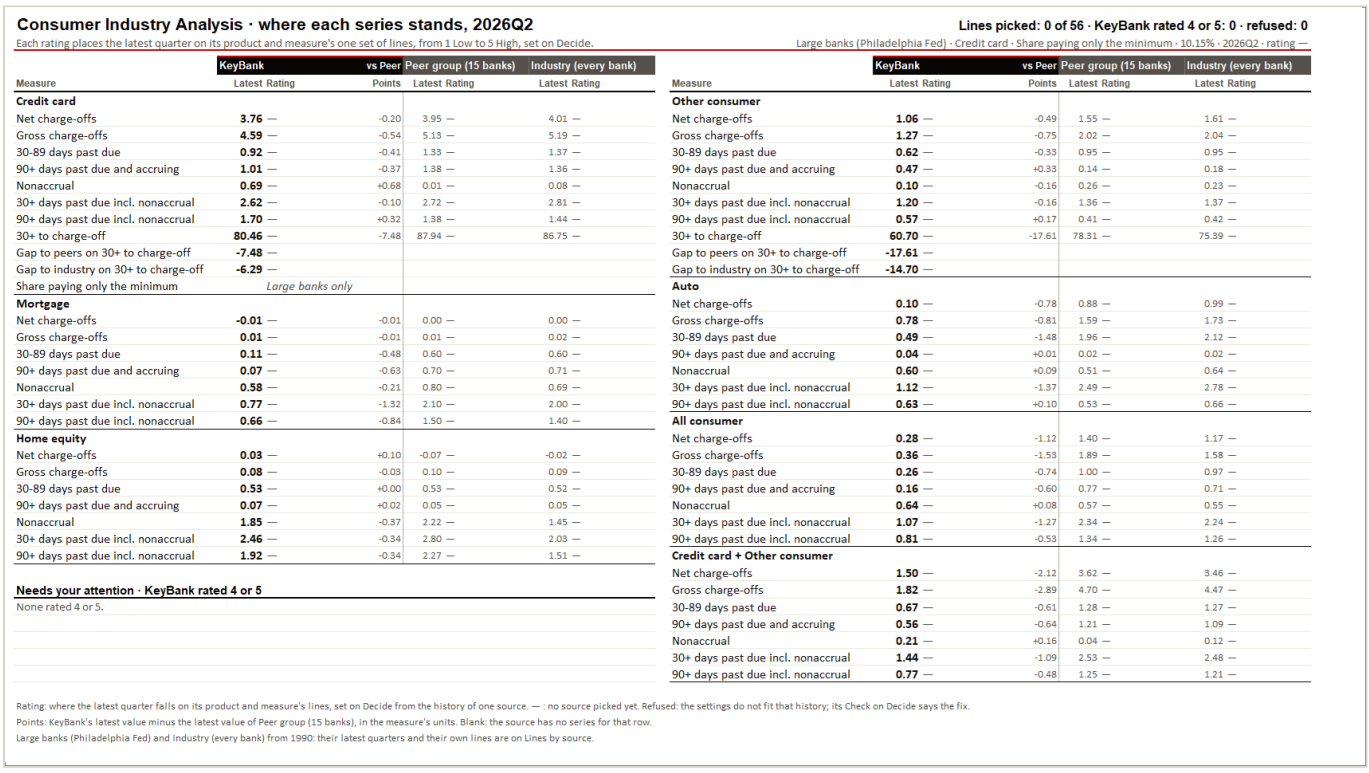
What the history suggests: keep every period, or leave one out?

Suggested	
Unusual stretches	
Why	
Left out now	No period is left out; every quarter is kept.

What each setting does when you raise it		What fits the history the lines are set from
Width of Moderate	More quarters rate 3 Moderate.	Pick a source under Lines set from to see what fits.
Moderate-Low begins	Fewer quarters rate 1 Low.	Pick a source under Lines set from to see what fits.
How rare High is (z)	Fewer quarters rate 5 High. A z of 2.5 is rare; 3.5 is the usual line for an outlier.	Pick a source under Lines set from to see what fits.
On a line, rate it	With worse, a value exactly on a line takes the worse rating.	Pick a source under Lines set from to see what fits.

The solid line is the source the lines are set from. A dashed line is another source switched on under On chart, in the color of its name. A gray stretch is a period left out of the history the lines are set from: it is drawn, and it does not set the lines.
The settings apply to every source's history: the lines picked, and what each source's own history would give. The shading runs from 1 Low to 5 High.
The chart starts at the first quarter any line has, within the window set under Chart shows. A diamond at the top of the shading marks a quarter in which a reporting change took effect (Sources tab, Reporting changes). Every term is defined on the Glossary tab.

B1 · Open the workbook as built (Decide tab, photographed by Excel during the walk)



Consumer Industry Analysis · where each series stands, 2026Q2

Lines picked: 0 of 56 · KeyBank rated 4 or 5: 0 · refused: 0

Each rating places the latest quarter on its product and measure's one set of lines, from 1 Low to 5 High, set on Decide.

Large banks (Philadelphia Fed) · Credit card · Share paying only the minimum · 10.15% · 2026Q2 · rating —

Measure	KeyBank Latest Rating	vs Peer Points	Peer group (15 banks) Latest Rating	Industry (every bank) Latest Rating
Credit card				
Net charge-offs	**3.76** —	-0.20	3.95 —	4.01 —
Gross charge-offs	**4.59** —	-0.54	5.13 —	5.19 —
30-89 days past due	**0.92** —	-0.41	1.33 —	1.37 —
90+ days past due and accruing	**1.01** —	-0.37	1.38 —	1.36 —
Nonaccrual	**0.69** —	+0.68	0.01 —	0.08 —
30+ days past due incl. nonaccrual	**2.62** —	-0.10	2.72 —	2.81 —
90+ days past due incl. nonaccrual	**1.70** —	+0.32	1.38 —	1.44 —
30+ to charge-off	**80.46** —	-7.48	87.94 —	86.75 —
Gap to peers on 30+ to charge-off	**-7.48** —			
Gap to industry on 30+ to charge-off	**-6.29** —			
Share paying only the minimum	*Large banks only*			
Mortgage				
Net charge-offs	**-0.01** —	-0.01	0.00 —	0.00 —
Gross charge-offs	**0.01** —	-0.01	0.01 —	0.02 —
30-89 days past due	**0.11** —	-0.48	0.60 —	0.60 —
90+ days past due and accruing	**0.07** —	-0.63	0.70 —	0.71 —
Nonaccrual	**0.58** —	-0.21	0.80 —	0.69 —
30+ days past due incl. nonaccrual	**0.77** —	-1.32	2.10 —	2.00 —
90+ days past due incl. nonaccrual	**0.66** —	-0.84	1.50 —	1.40 —
Home equity				
Net charge-offs	**0.03** —	+0.10	-0.07 —	-0.02 —
Gross charge-offs	**0.08** —	-0.03	0.10 —	0.09 —
30-89 days past due	**0.53** —	+0.00	0.53 —	0.52 —
90+ days past due and accruing	**0.07** —	+0.02	0.05 —	0.05 —
Nonaccrual	**1.85** —	-0.37	2.22 —	1.45 —
30+ days past due incl. nonaccrual	**2.46** —	-0.34	2.80 —	2.03 —
90+ days past due incl. nonaccrual	**1.92** —	-0.34	2.27 —	1.51 —

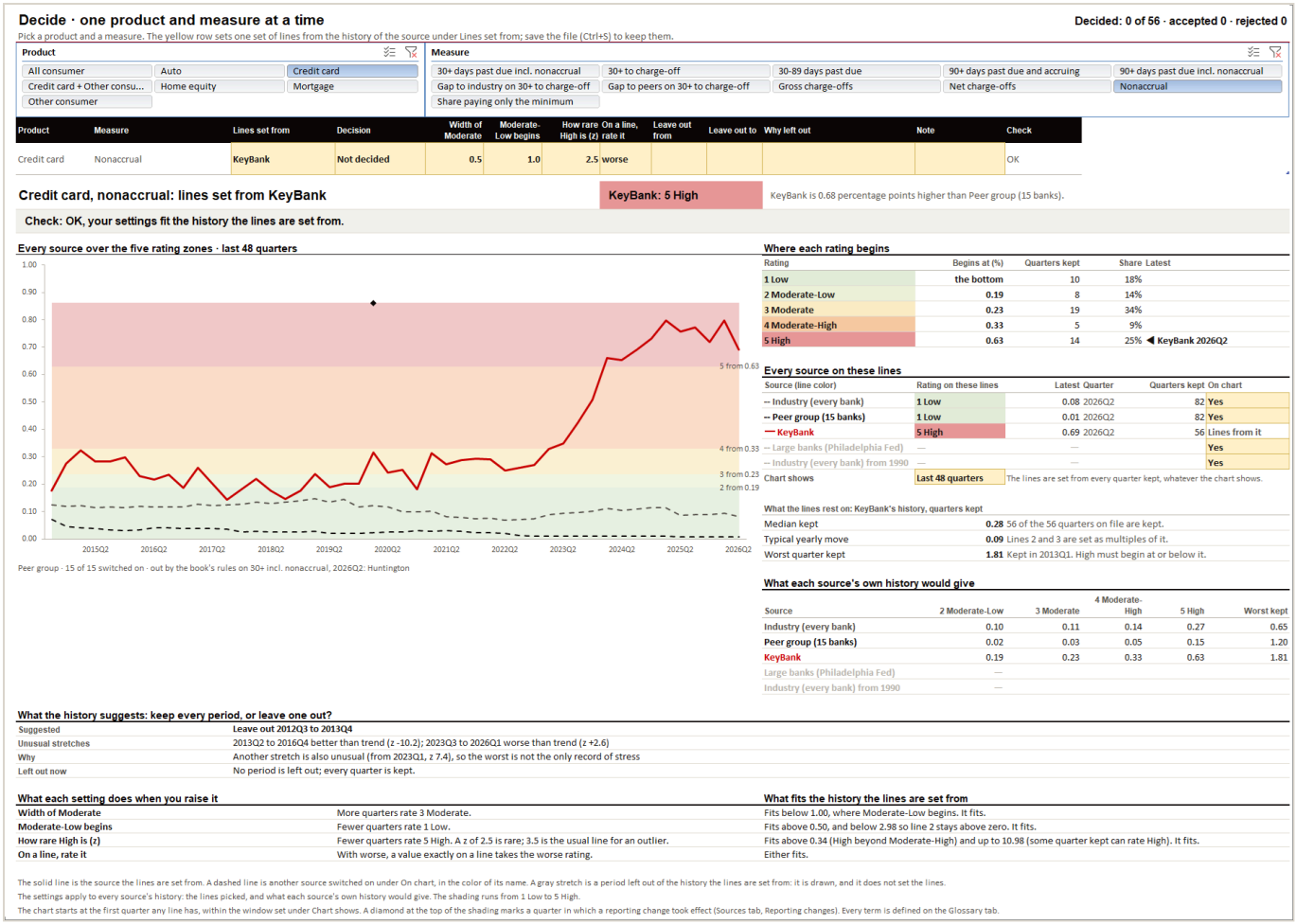
Needs your attention · KeyBank rated 4 or 5

None rated 4 or 5.

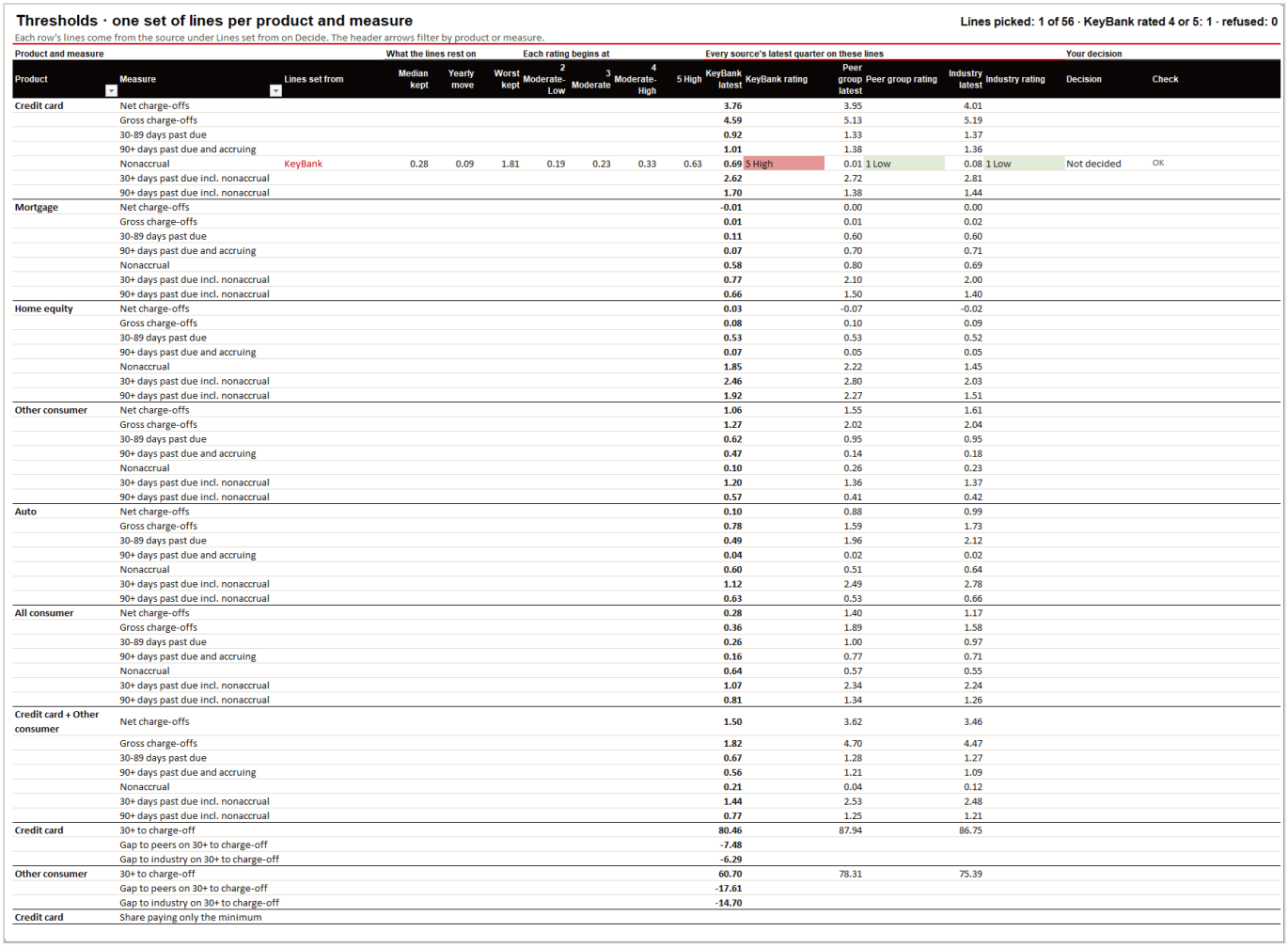
Measure	KeyBank Latest Rating	vs Peer Points	Peer group (15 banks) Latest Rating	Industry (every bank) Latest Rating
Other consumer				
Net charge-offs	**1.06** —	-0.49	1.55 —	1.61 —
Gross charge-offs	**1.27** —	-0.75	2.02 —	2.04 —
30-89 days past due	**0.62** —	-0.33	0.95 —	0.95 —
90+ days past due and accruing	**0.47** —	+0.33	0.14 —	0.18 —
Nonaccrual	**0.10** —	-0.16	0.26 —	0.23 —
30+ days past due incl. nonaccrual	**1.20** —	-0.16	1.36 —	1.37 —
90+ days past due incl. nonaccrual	**0.57** —	+0.17	0.41 —	0.42 —
30+ to charge-off	**60.70** —	-17.61	78.31 —	75.39 —
Gap to peers on 30+ to charge-off	**-17.61** —			
Gap to industry on 30+ to charge-off	**-14.70** —			
Auto				
Net charge-offs	**0.10** —	-0.78	0.88 —	0.99 —
Gross charge-offs	**0.78** —	-0.81	1.59 —	1.73 —
30-89 days past due	**0.49** —	-1.48	1.96 —	2.12 —
90+ days past due and accruing	**0.04** —	+0.01	0.02 —	0.02 —
Nonaccrual	**0.60** —	+0.09	0.51 —	0.64 —
30+ days past due incl. nonaccrual	**1.12** —	-1.37	2.49 —	2.78 —
90+ days past due incl. nonaccrual	**0.63** —	+0.10	0.53 —	0.66 —
All consumer				
Net charge-offs	**0.28** —	-1.12	1.40 —	1.17 —
Gross charge-offs	**0.36** —	-1.53	1.89 —	1.58 —
30-89 days past due	**0.26** —	-0.74	1.00 —	0.97 —
90+ days past due and accruing	**0.16** —	-0.60	0.77 —	0.71 —
Nonaccrual	**0.64** —	+0.08	0.57 —	0.55 —
30+ days past due incl. nonaccrual	**1.07** —	-1.27	2.34 —	2.24 —
90+ days past due incl. nonaccrual	**0.81** —	-0.53	1.34 —	1.26 —
Credit card + Other consumer				
Net charge-offs	**1.50** —	-2.12	3.62 —	3.46 —
Gross charge-offs	**1.82** —	-2.89	4.70 —	4.47 —
30-89 days past due	**0.67** —	-0.61	1.28 —	1.27 —
90+ days past due and accruing	**0.56** —	-0.64	1.21 —	1.09 —
Nonaccrual	**0.21** —	+0.16	0.04 —	0.12 —
30+ days past due incl. nonaccrual	**1.44** —	-1.09	2.53 —	2.48 —
90+ days past due incl. nonaccrual	**0.77** —	-0.48	1.25 —	1.21 —

Rating: where the latest quarter falls on its product and measure's lines, set on Decide from the history of one source. — : no source picked yet. Refused: the settings do not fit that history; its Check on Decide says the fix.
Points: KeyBank's latest value minus the latest value of Peer group (15 banks), in the measure's units. Blank: the source has no series for that row.
Large banks (Philadelphia Fed) and Industry (every bank) from 1990: their latest quarters and their own lines are on Lines by source.

B1 · Open the workbook as built (Summary tab, photographed by Excel during the walk)



B3 · On Decide, click Credit card and Nonaccrual, and pick KeyBank under Lines set from (Decide tab, photographed by Excel during the walk)



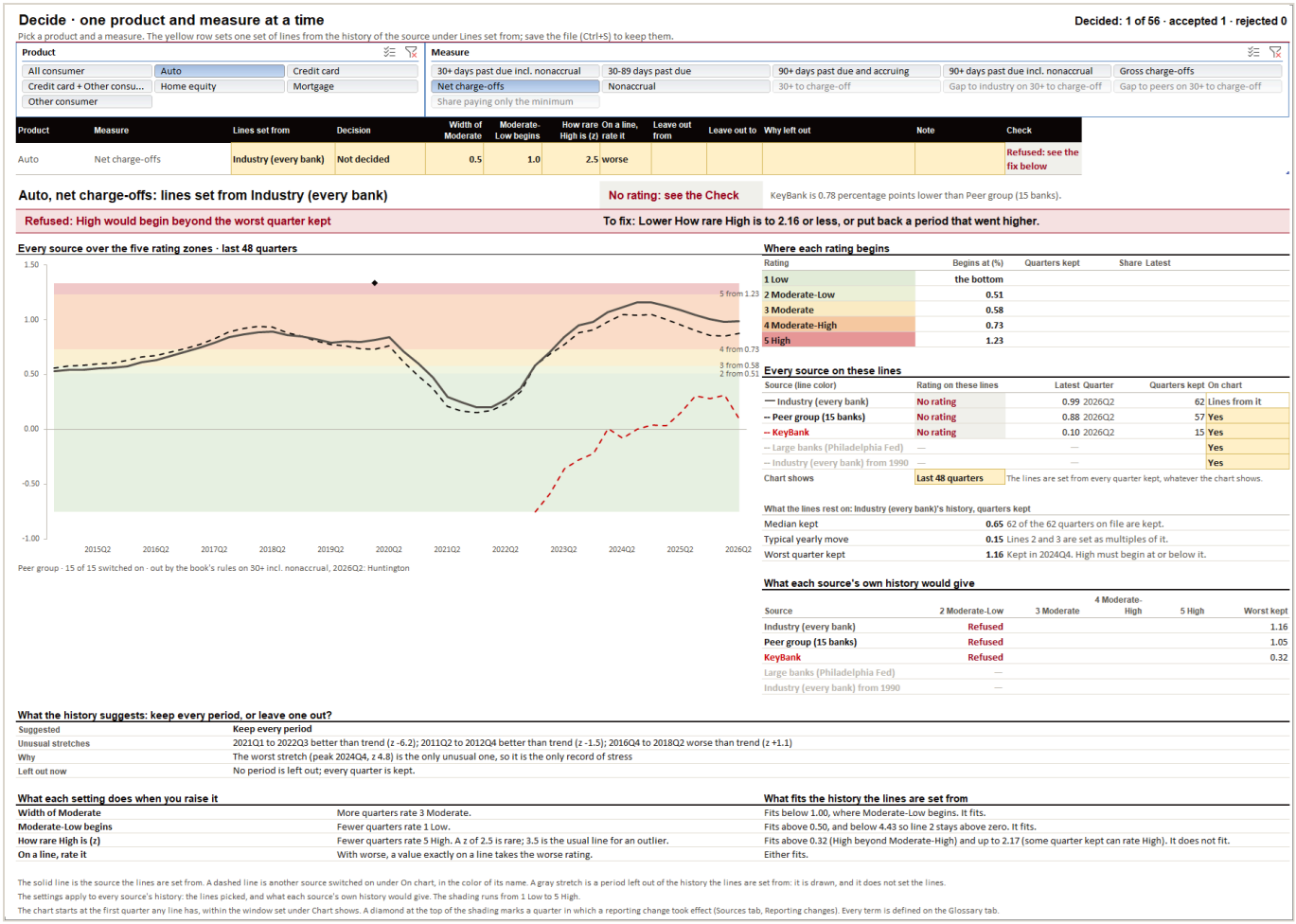
Thresholds · one set of lines per product and measure

Lines picked: 1 of 56 · KeyBank rated 4 or 5: 1 · refused: 0

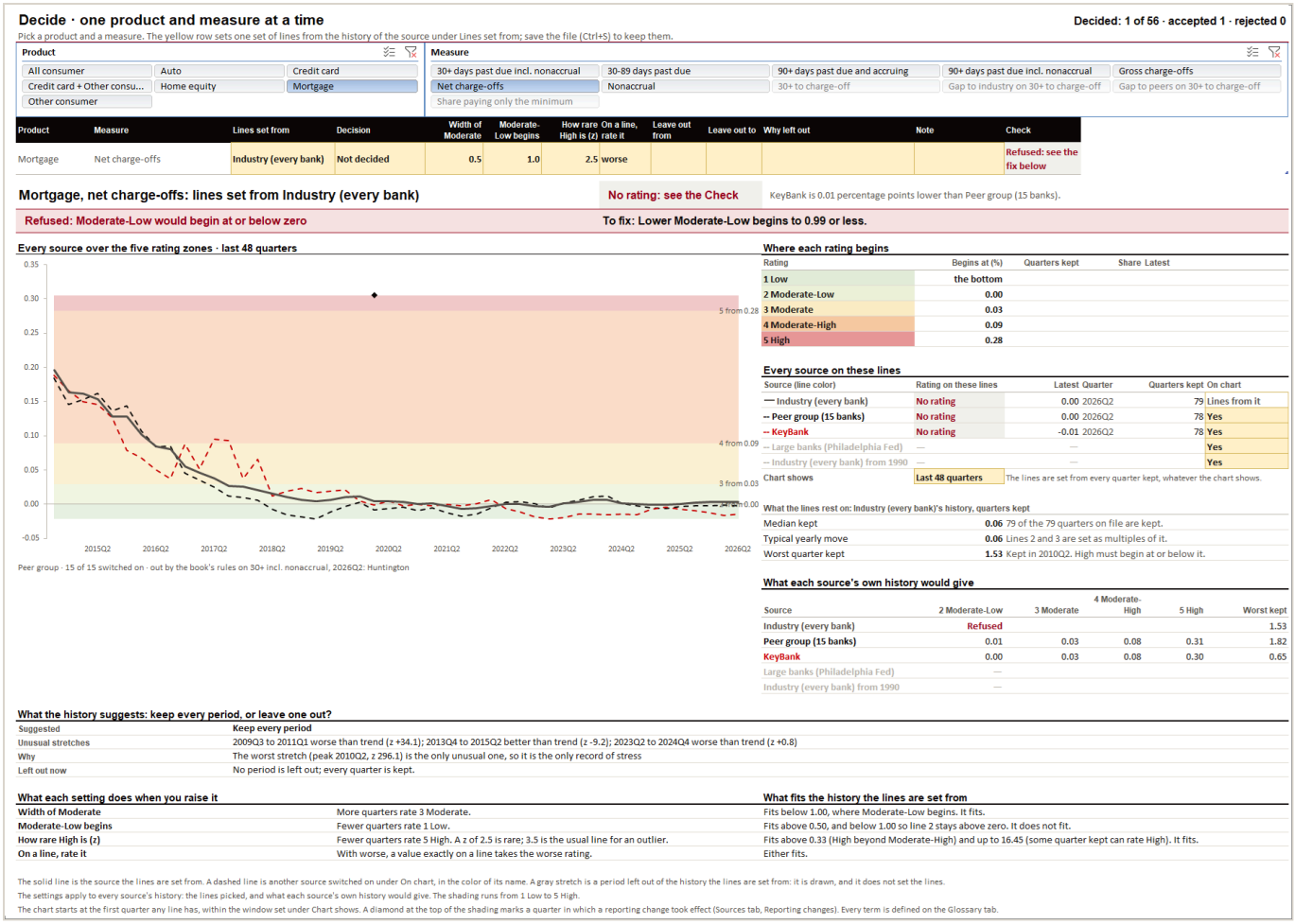
Each row's lines come from the source under Lines set from on Decide. The header arrows filter by product or measure.

Product and measure			What the lines rest on		Each rating begins at					Every source's latest quarter on these lines					Your decision	
Product	Measure	Lines set from	Median kept	Yearly move	Worst kept	2 Moderate-Low	3 Moderate	4 Moderate-High	5 High	KeyBank latest	KeyBank rating	Peer group latest	Peer group rating	Industry latest	Industry rating	Decision	Check
Credit card	Net charge-offs									**3.76**		3.95		4.01			
	Gross charge-offs									**4.59**		5.13		5.19			
	30-89 days past due									**0.92**		1.33		1.37			
	90+ days past due and accruing									**1.01**		1.38		1.36			
	Nonaccrual	KeyBank	0.28	0.09	1.81	0.19	0.23	0.33	0.63	**0.69**	5 High	0.01	1 Low	0.08	1 Low	Not decided	OK
	30+ days past due incl. nonaccrual									**2.62**		2.72		2.81			
	90+ days past due incl. nonaccrual									**1.70**		1.38		1.44			
Mortgage	Net charge-offs									**-0.01**		0.00		0.00			
	Gross charge-offs									**0.01**		0.01		0.02			
	30-89 days past due									**0.11**		0.60		0.60			
	90+ days past due and accruing									**0.07**		0.70		0.71			
	Nonaccrual									**0.58**		0.80		0.69			
	30+ days past due incl. nonaccrual									**0.77**		2.10		2.00			
	90+ days past due incl. nonaccrual									**0.66**		1.50		1.40			
Home equity	Net charge-offs									**0.03**		-0.07		-0.02			
	Gross charge-offs									**0.08**		0.10		0.09			
	30-89 days past due									**0.53**		0.53		0.52			
	90+ days past due and accruing									**0.07**		0.05		0.05			
	Nonaccrual									**1.85**		2.22		1.45			
	30+ days past due incl. nonaccrual									**2.46**		2.80		2.03			
	90+ days past due incl. nonaccrual									**1.92**		2.27		1.51			
Other consumer	Net charge-offs									**1.06**		1.55		1.61			
	Gross charge-offs									**1.27**		2.02		2.04			
	30-89 days past due									**0.62**		0.95		0.95			
	90+ days past due and accruing									**0.47**		0.14		0.18			
	Nonaccrual									**0.10**		0.26		0.23			
	30+ days past due incl. nonaccrual									**1.20**		1.36		1.37			
	90+ days past due incl. nonaccrual									**0.57**		0.41		0.42			
Auto	Net charge-offs									**0.10**		0.88		0.99			
	Gross charge-offs									**0.78**		1.59		1.73			
	30-89 days past due									**0.49**		1.96		2.12			
	90+ days past due and accruing									**0.04**		0.02		0.02			
	Nonaccrual									**0.60**		0.51		0.64			
	30+ days past due incl. nonaccrual									**1.12**		2.49		2.78			
	90+ days past due incl. nonaccrual									**0.63**		0.53		0.66			
All consumer	Net charge-offs									**0.28**		1.40		1.17			
	Gross charge-offs									**0.36**		1.89		1.58			
	30-89 days past due									**0.26**		1.00		0.97			
	90+ days past due and accruing									**0.16**		0.77		0.71			
	Nonaccrual									**0.64**		0.57		0.55			
	30+ days past due incl. nonaccrual									**1.07**		2.34		2.24			
	90+ days past due incl. nonaccrual									**0.81**		1.34		1.26			
Credit card + Other consumer	Net charge-offs									**1.50**		3.62		3.46			
	Gross charge-offs									**1.82**		4.70		4.47			
	30-89 days past due									**0.67**		1.28		1.27			
	90+ days past due and accruing									**0.56**		1.21		1.09			
	Nonaccrual									**0.21**		0.04		0.12			
	30+ days past due incl. nonaccrual									**1.44**		2.53		2.48			
	90+ days past due incl. nonaccrual									**0.77**		1.25		1.21			
Credit card	30+ to charge-off									**80.46**		87.94		86.75			
	Gap to peers on 30+ to charge-off									**-7.48**							
	Gap to industry on 30+ to charge-off									**-6.29**							
Other consumer	30+ to charge-off									**60.70**		78.31		75.39			
	Gap to peers on 30+ to charge-off									**-17.61**							
	Gap to industry on 30+ to charge-off									**-14.70**							
Credit card	Share paying only the minimum																

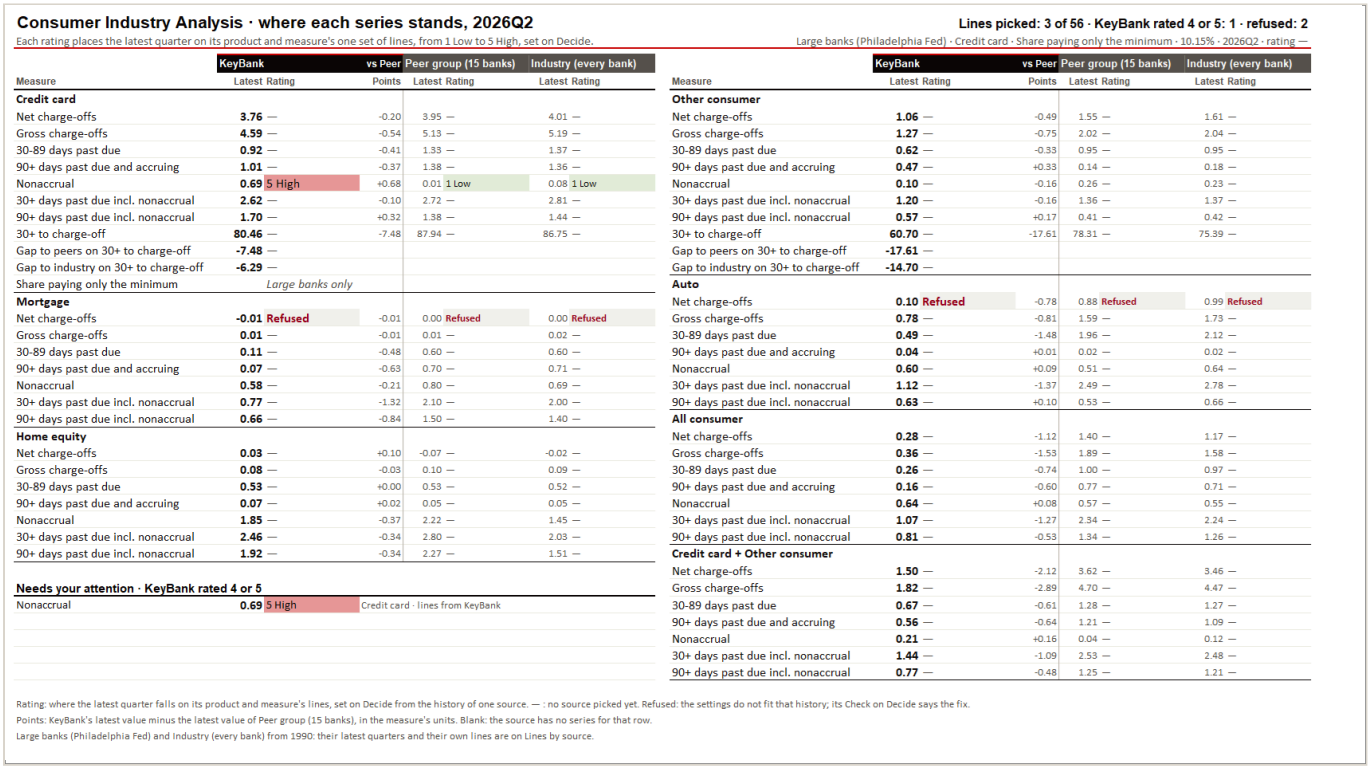
B3 · On Decide, click Credit card and Nonaccrual, and pick KeyBank under Lines set from (Thresholds tab, photographed by Excel during the walk)



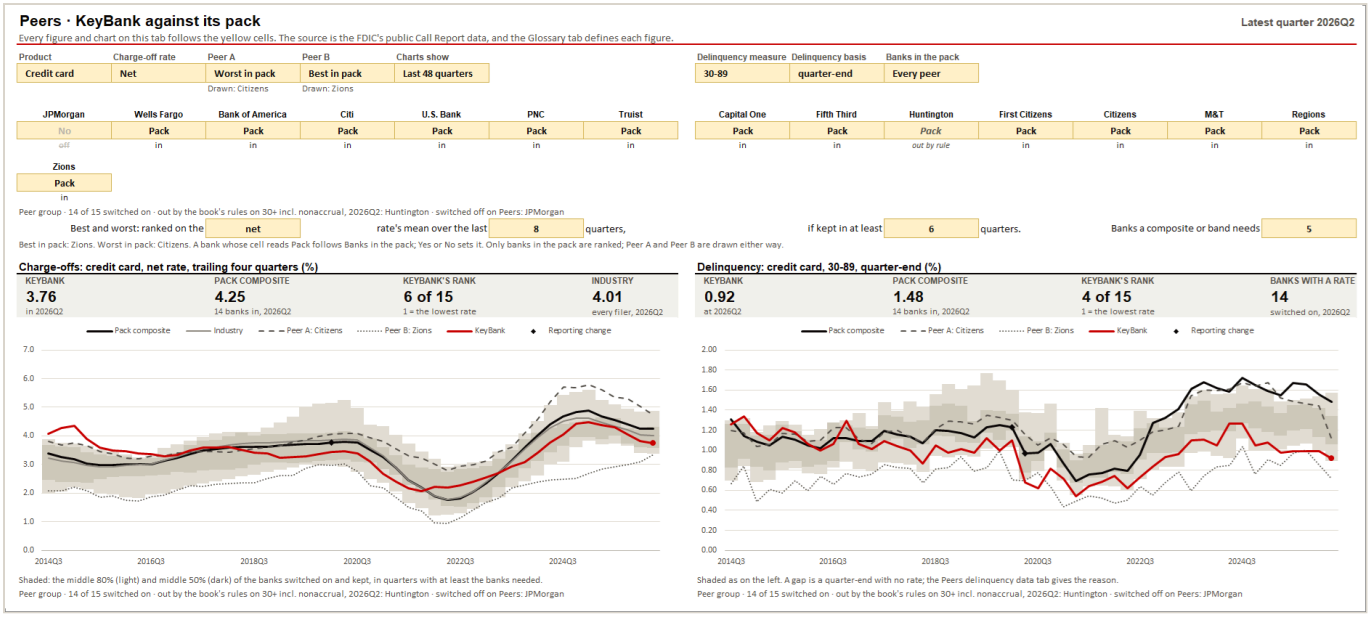
B8 · Click Auto and Net charge-offs, and pick Industry (every bank) under Lines set from (Decide tab, photographed by Excel during the walk)



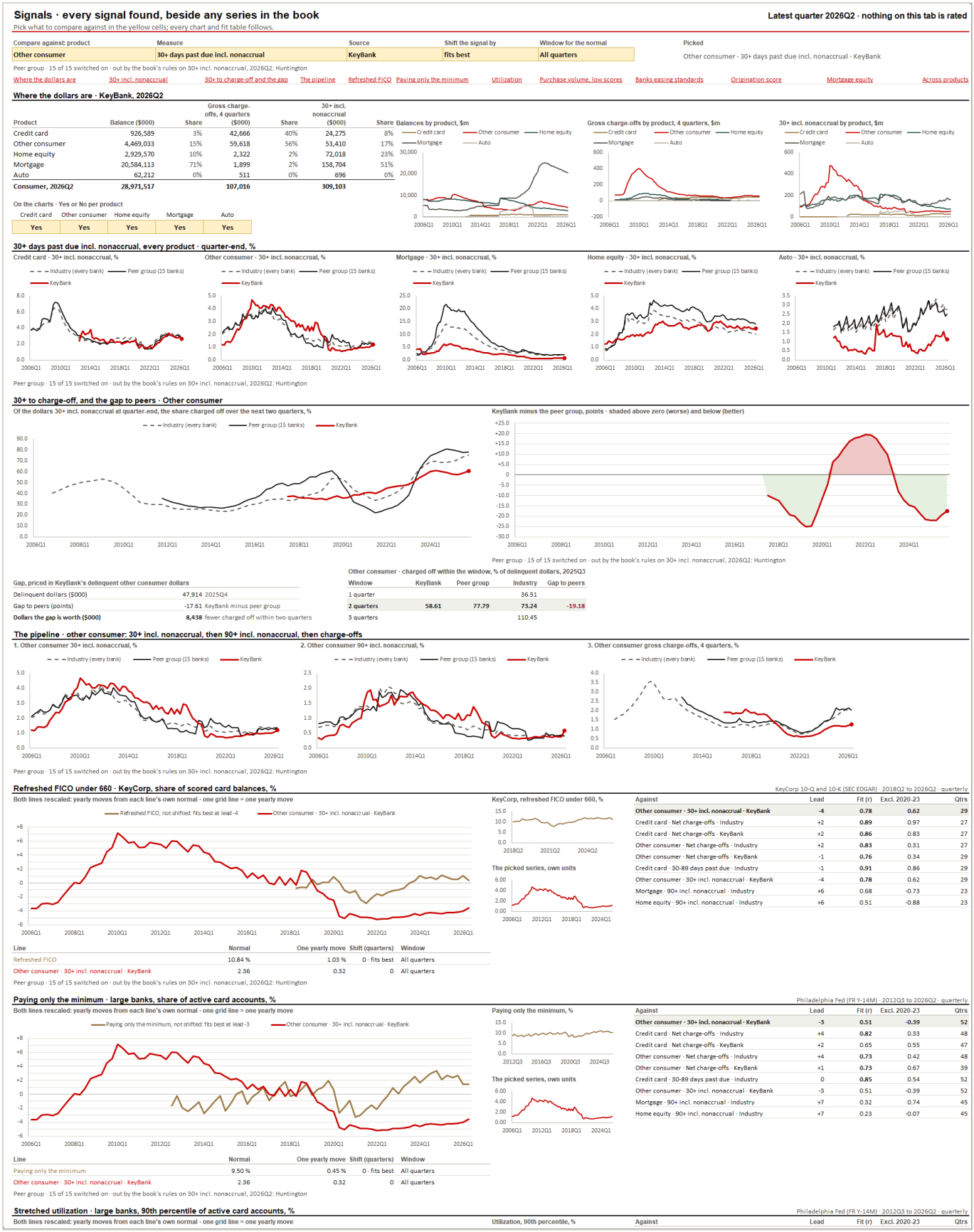
B9 · Click Mortgage and Net charge-offs, and pick Industry (every bank) (Decide tab, photographed by Excel during the walk)



B10 · Save (Ctrl+S), close Excel, open the file again (Summary tab, photographed by Excel during the walk)



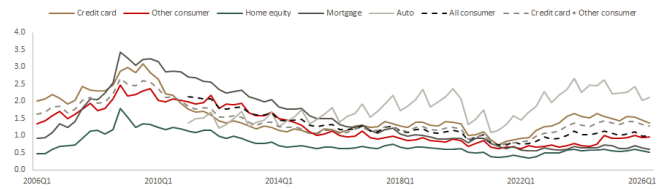
B12 · Set Banks in the pack back to Every peer, and JPMorgan's cell to No (Peers tab, photographed by Excel during the walk)



B14 · Open Signals (Signals tab, photographed by Excel during the walk)

Across products · 30-89 days past due, Industry (every bank), against the picked series

30-89 days past due, every product, Industry (every bank), %



Against Other consumer · 30+ incl. nonaccrual · KeyBank

Product	Lead	Fit (r)	Excl. 2020-23	Qtrs
Credit card	+3	0.41	0.47	75
Other consumer	+3	0.66	0.71	75
Home equity	+5	**0.73**	0.81	73
Mortgage	+3	**0.73**	0.79	75
Auto	-3	0.51	-0.26	58
All consumer	-3	0.42	0.18	58
Credit card + Other consumer	+3	0.52	0.61	75

B14 · Open Signals (Signals tab, photographed by Excel during the walk)